



CSE303: Statistics for Data Science
Spring 2026
Group- 6

“Modeling the Relationship Between Workload and Mental Health in University Students”

Term Project Report

[Project Link](#)

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1. Introduction

University students regularly face academic challenges, heavy workloads, deadlines, and extracurricular activities, which often result in increased stress. It is widely acknowledged that these factors can negatively affect their mental wellbeing. To obtain more evidence-based insights, quantitatively examining how specific factors contribute to variations in stress, anxiety, and sleep is necessary. This project aims to explore the relationship between academic workload and mental health indicators through a data-driven approach. A survey was designed to collect relevant key variables which include weekly study hours, enrolled courses, sleep patterns, stress levels and anxiety scores. Prior to analysis, the collected dataset was then systematically preprocessed to ensure consistency.

Exploratory data analysis was conducted to characterize the dataset and identify underlying patterns using statistical methods of measure such as mean, dispersion, correlation. Simple linear regression models were used to quantify the relationship between workload variables and mental wellbeing ones.

The findings from this study aim to provide empirical insights into how workload impacts student well-being. The analysis is focused on identifying measurable relationships between workload variables and mental health indicators through statistical and regression-based methods.

2. Survey Questions:



The survey had 24 questions asking for a student's age, semester, CGPA, department, stress level and more like:

“How many hours do you study per week?”

“How many assignments/projects/exams do you handle per week?”

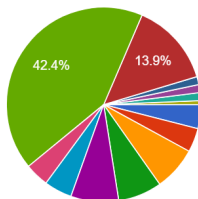
“How often do you feel overwhelmed by academic workload?”

“Do you have a part-time job?”

We got 151 responses on our survey.

Semester

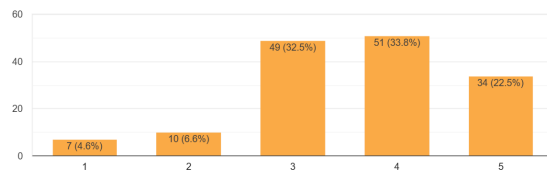
151 responses



How often do you feel overwhelmed by academic workload?

151 responses

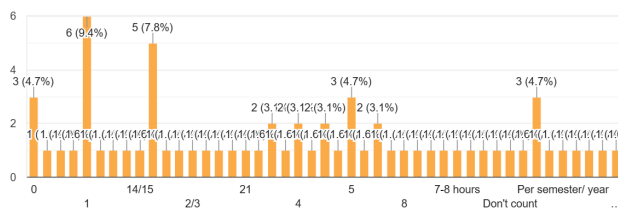
[Copy chart](#)



If yes, how many hours per week?

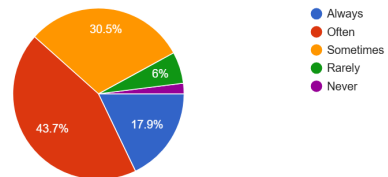
64 responses

[Copy chart](#)



How often do you feel mentally exhausted?

151 responses



3. Dataset Manipulation

The number of responses that we got from our survey was not enough so we had to expand the original dataset up-to around 1000 rows. We used LLM to do that.

4. Data Preprocessing

The dataset was preprocessed systematically to ensure accuracy, consistency, and suitability for statistical analysis and modeling.

a. Data Cleaning:

The dataset was initially examined for missing values, duplicate entries, and inconsistencies in formatting. A small number of missing values were identified across variables such as CGPA, Courses, StudyHours, and SleepHours. These were handled appropriately to maintain data integrity. Additionally, one duplicate record was detected and removed.

b. Feature Selection and Renaming:

Relevant variables related to academic workload, lifestyle, and mental health were selected from the original dataset. Column names were standardized for clarity and consistency (e.g., StudyHours, PartTimeHours, ExtraCurricularHours), enabling easier interpretation and analysis.

Cleaned Data Shape: (998, 11)						
Cleaned Data Preview:						
	Year	Gender	CGPA	Courses	StudyHours	PartTimeHours \
0	6	1	3.5	4.0	8.0	8.0
1	4	0	3.5	5.0	17.0	6.0
2	5	0	2.5	3.0	8.0	0.0
3	2	1	3.5	3.0	40.0	14.0
4	8	0	2.5	4.0	30.0	20.0
	ExtraCurricularHours		Stress	Anxiety	SleepQuality	SleepHours
0			2.0	10.0	9.0	1.0 7.0
1			0.0	2.0	2.0	4.0 5.0
2			0.0	5.0	1.0	9.0 6.0
3			0.0	6.0	7.0	6.0 7.0
4			0.0	5.0	3.0	2.0 8.0

c. Feature Identification:

Variables were categorized based on their data types:

1. Numerical features: CGPA, StudyHours, Courses, SleepHours, Stress, Anxiety
2. Categorical features: Gender, Year of study
3. Ordinal features: SleepQuality

d. Numeric Value Standardization:

A custom preprocessing function was developed to handle inconsistent and non-uniform numeric inputs. Converting range-based responses (e.g., “5–6”) into their average values, extracting numeric values from mixed text inputs, converting all valid entries into consistent floating-point format.

This step ensured uniformity across all numerical variables and reduced data noise.

e. Encoding of Categorical Variables:

Categorical variables were converted into numerical format to support statistical analysis. Gender was encoded using binary encoding: Male = 0, Female = 1

f. Handling Missing Values:

Missing values were treated using both statistical and logical imputation methods:

Median imputation was applied to CGPA, Courses, StudyHours, and SleepHours to preserve distribution robustness. Missing values in PartTimeHours and Extra CurricularHours were replaced with 0, based on the assumption that absence of response indicates no participation

Missing Before:		Missing After:	
Year	0	Year	0
Gender	0	Gender	0
CGPA	7	CGPA	0
Courses	1	Courses	0
StudyHours	4	StudyHours	0
PartTimeHours	0	PartTimeHours	0
ExtraCurricularHours	0	ExtraCurricularHours	0
Stress	0	Stress	0
Anxiety	0	Anxiety	0
SleepQuality	0	SleepQuality	0
SleepHours	1	SleepHours	0
dtype: int64		dtype: int64	
[Imputation] CGPA: filled 7 nulls with median = 3.00			
[Imputation] Courses: filled 1 nulls with median = 4.00			
[Imputation] StudyHours: filled 4 nulls with median = 12.00			
[Imputation] SleepHours: filled 1 nulls with median = 6.00			

g. Exploration of Measurement Scales:

The dataset included variables measured on different scales:

1. Stress and Anxiety: 1–5 Likert scale
2. SleepQuality: ordinal rating scale
3. Continuous variables: hours and CGPA

h. Outlier Detection and Handling:

Outliers were addressed using rule-based filtering grounded in realistic constraints:

1. StudyHours values greater than 100 hours per week were removed
2. SleepHours values exceeding 24 hours per day were excluded

i. Normalization and Scaling:

Min–Max normalization was applied to selected continuous variables (StudyHours, Courses, SleepHours, CGPA, PartTimeHours, ExtraCurricularHours) to standardize their ranges for comparative analysis.

```
[Normalization] StudyHours: min=0.00, max=42.00  
[Normalization] Courses: min=2.00, max=24.00  
[Normalization] SleepHours: min=0.00, max=10.00  
[Normalization] CGPA: min=2.50, max=3.50  
[Normalization] PartTimeHours: min=0.00, max=168.00  
[Normalization] ExtraCurricularHours: min=0.00, max=42.00
```

j. Assumption Preparation for Modeling:

Preprocessing ensured that the dataset satisfied key requirements for regression analysis. Clean numerical inputs, consistent formatting, and removal of anomalies enabled further evaluation of model assumptions such as linearity and error distribution.

k. Final Dataset Validation:

The processed dataset was validated to ensure consistency with expected ranges and survey design. After preprocessing:

- No missing values remained
- Duplicate entries were removed
- All variables were properly formatted

5. Dataset Characteristics

The dataset consists of 998 complete student responses, structured across 11 carefully selected variables that capture academic workload, lifestyle behaviors, and mental health indicators.

● Feature Composition

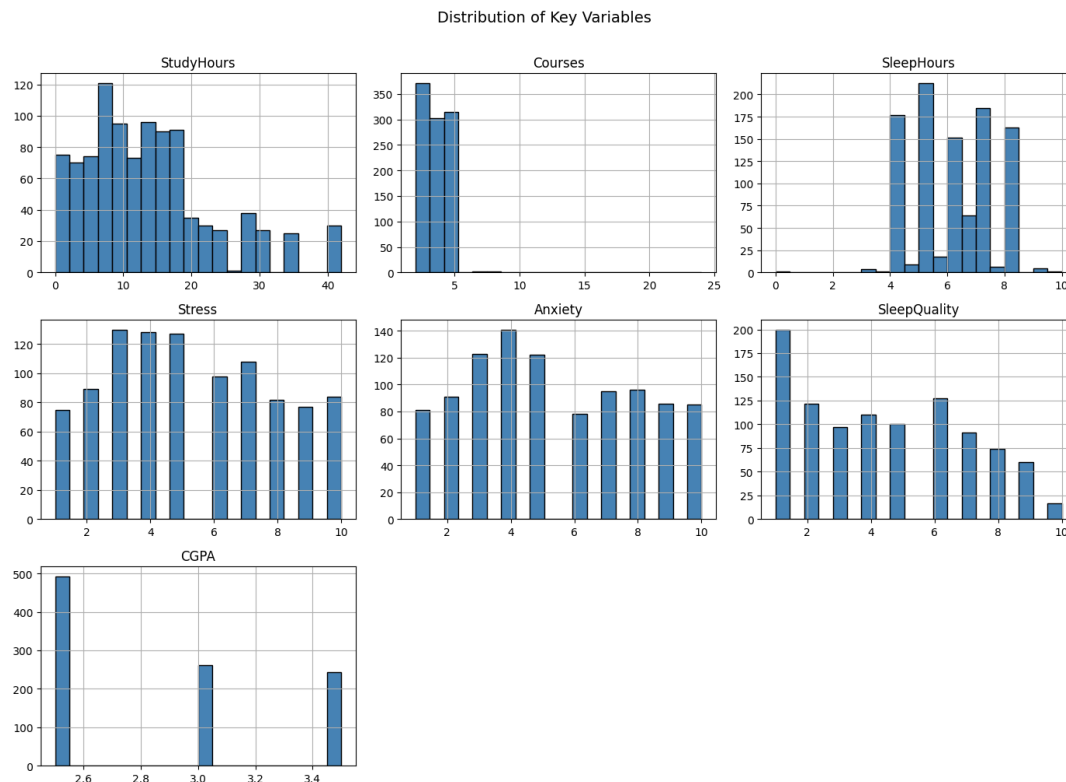
The dataset includes a balanced combination of variables:

- Academic variables: CGPA, Courses, StudyHours
- Mental health indicators: Stress, Anxiety
- Lifestyle factors: SleepHours, SleepQuality, PartTimeHours, ExtraCurricularHours
- Demographic variables: Gender, Year

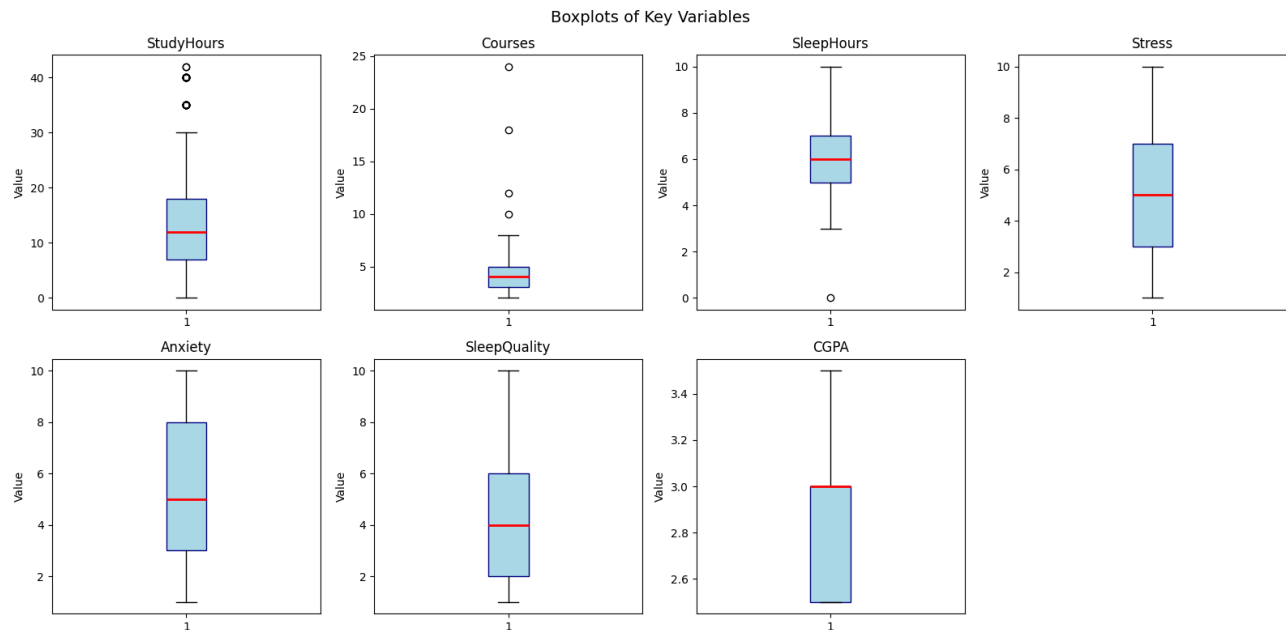
6. Exploratory Data Analysis – Statistical Analysis

Exploratory Data Analysis was conducted to understand the structure, distribution, and relationships within the dataset before applying regression modeling. The analysis combined both numerical statistical measures and graphical visualizations to extract meaningful patterns from the data.

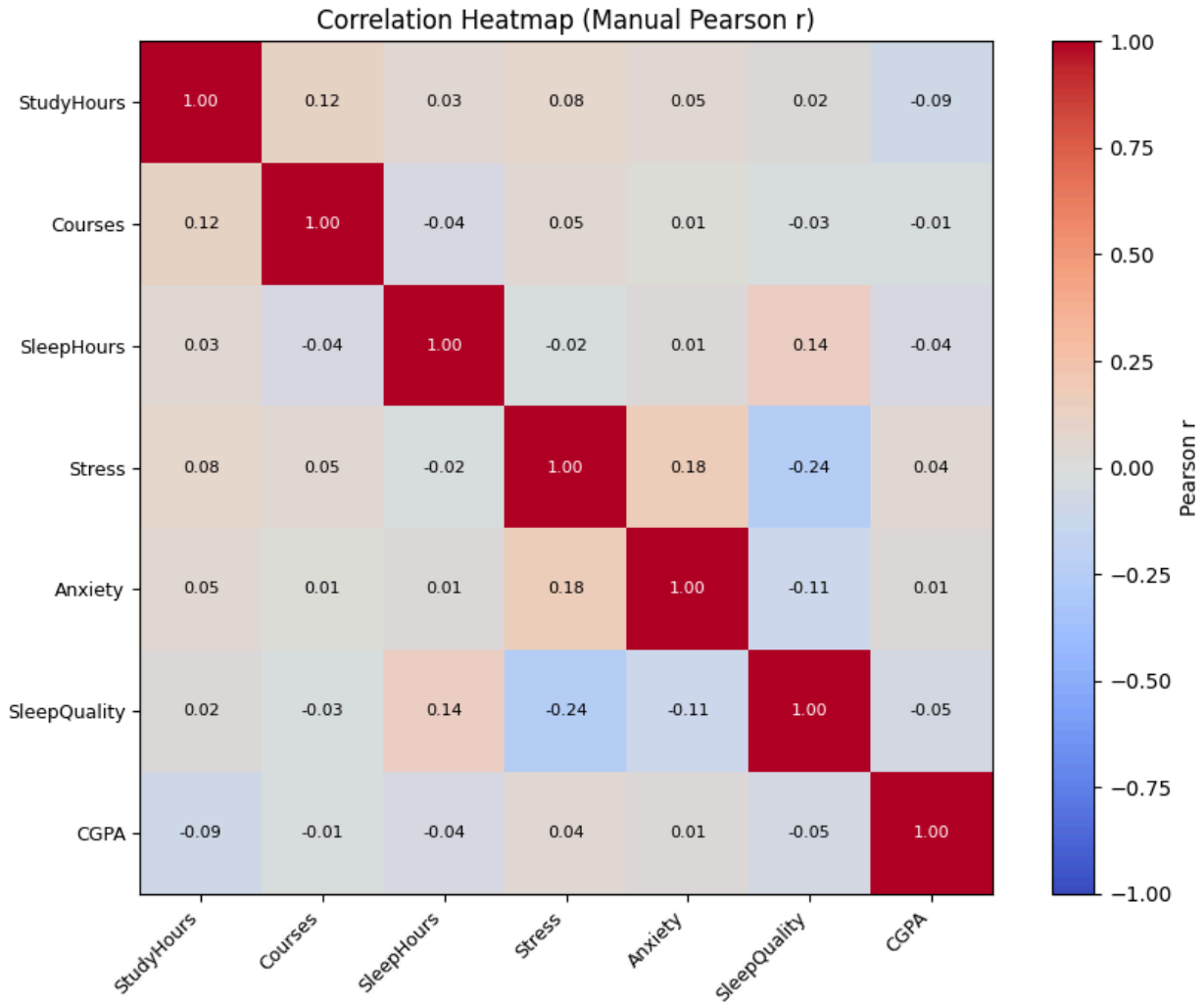
Descriptive statistics such as mean, median, variance, standard deviation, quartiles, and interquartile range (IQR) were computed for key variables including study hours, number of courses, sleep hours, stress, anxiety, and CGPA. These measures provided a summary of central tendency and variability, highlighting differences in academic workload and mental health indicators among students. To visually analyze the data, histograms were generated for all key variables. These histograms helped in understanding the distribution patterns, showing whether variables such as stress and study hours were evenly distributed or concentrated within specific ranges.



Additionally, skewness values were calculated to assess the symmetry of distributions, revealing whether the data was normally distributed or skewed. Boxplots were also used to detect variability and potential outliers in the dataset, providing a clear view of spread and extreme values.



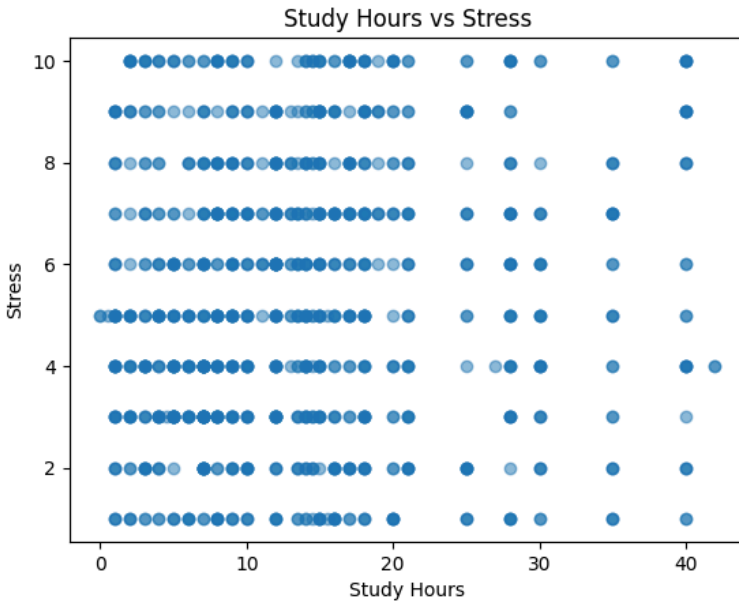
Scatter plots were used to visually examine relationships between key variable pairs such as study hours vs stress, courses vs anxiety, and sleep hours vs stress. These plots helped identify linear trends that later justified the use of linear regression modeling.



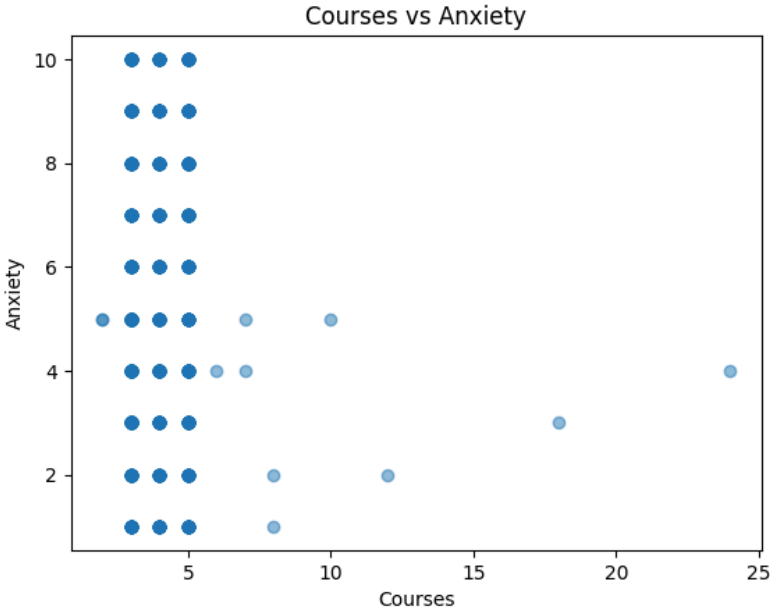
Study hours have a very weak positive correlation with stress ($r \approx 0.08$) and anxiety ($r \approx 0.05$), suggesting minimal influence. Similarly, sleep hours show an almost negligible negative correlation with stress ($r \approx -0.02$). The number of courses also has very weak correlations with both stress and anxiety.

The strongest observed relationship is between stress and sleep quality ($r \approx -0.24$), indicating that higher stress is moderately associated with poorer sleep quality. Additionally, stress and anxiety show a weak positive relationship ($r \approx 0.18$), suggesting they tend to increase together.

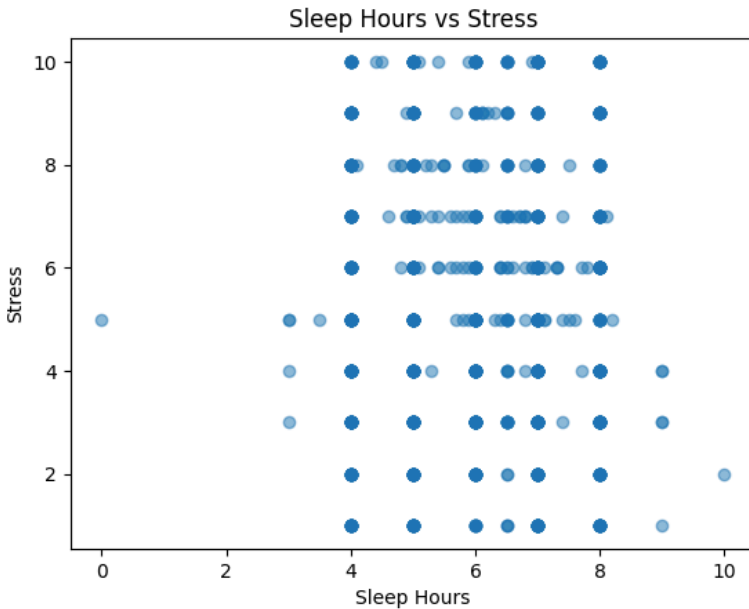
Overall, the heatmap indicates that no strong linear relationships exist among the variables, supporting the later regression results that these factors alone are insufficient to explain variations in student mental health.



This scatter plot illustrates the relationship between weekly study hours and stress levels among students. The data points show a generally weak to moderate positive relationship, where an increase in study hours is often associated with higher stress levels. The relationship is not strongly linear as stress levels remain widely distributed across all ranges of study hours. This indicates that while academic workload contributes to stress, it is not the sole determining factor.



This plot shows the relationship between the number of courses enrolled and anxiety levels. The distribution indicates a very weak and scattered relationship, with no strong linear pattern observed. Most students are concentrated around a similar number of courses, while anxiety levels vary significantly within that range. A few outliers with higher course loads show slightly higher anxiety, but this trend is not consistent across the dataset. This suggests that the number of courses alone may not be a strong predictor of anxiety.



This scatter plot represents the relationship between sleep duration and stress levels. A clear negative trend can be observed, where students with fewer sleep hours tend to report higher stress levels. Although the relationship is not perfectly linear, the overall pattern suggests that reduced sleep is associated with increased stress. Students with around 4–6 hours of sleep generally show higher stress scores compared to those with 7–9 hours of sleep. This inverse relationship highlights sleep duration as an important factor influencing mental well-being. Compared to the previous plots, this relationship appears relatively stronger and more consistent.

As part of the descriptive analysis, multiple academic, behavioral, and psychological variables were analyzed across the dataset. Overall, study hours show relatively high variability (mean = 13.74, std = 9.28). This indicates the significant differences in student study patterns. This variability is consistent with the previously visualized histogram and boxplot, where a wide spread and potential extreme values were observed. The number of courses taken per student is relatively stable (mean = 4.01, std = 1.20), although it exhibits strong positive skewness (6.66), suggesting the presence of a small number of students taking unusually high course loads.

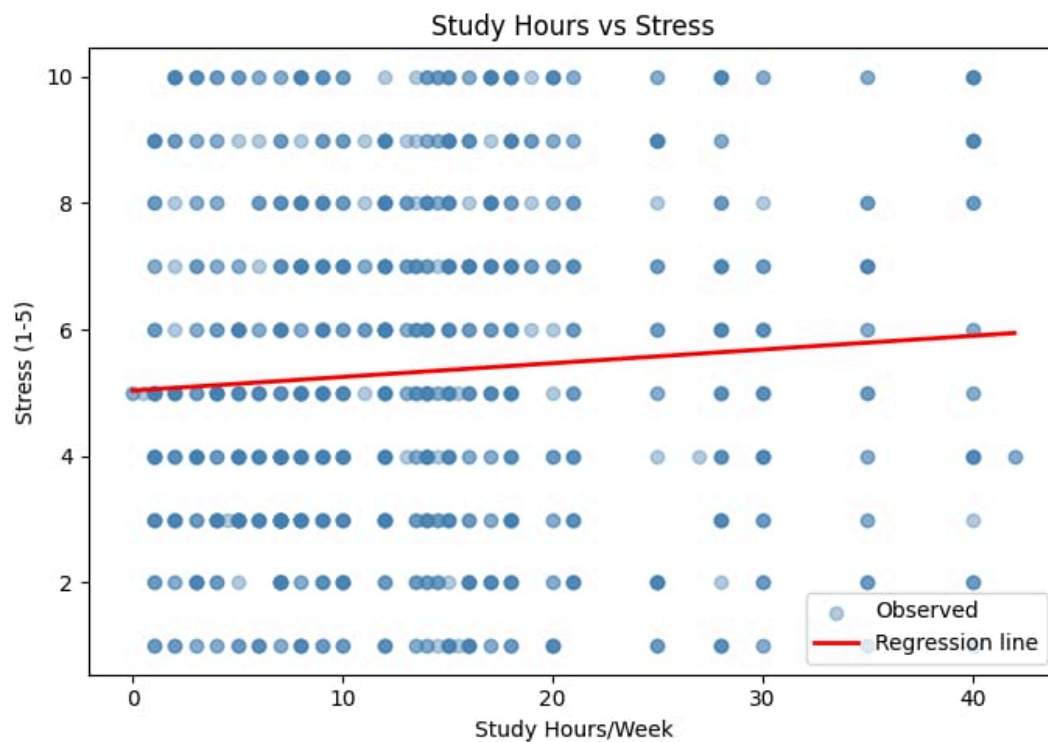
In contrast, sleep hours are more normally distributed (mean = 5.98, skewness ≈ 0), indicating a balanced spread across students corresponding to the histogram visualization. For psychological variables, both stress (mean = 5.33) and anxiety (mean = 5.34) show moderate variability, suggesting heterogeneous mental health conditions across students. CGPA (mean = 2.88, std = 0.41) demonstrates relatively low variability, indicating that academic performance is more concentrated around the average compared to behavioral or psychological variables.

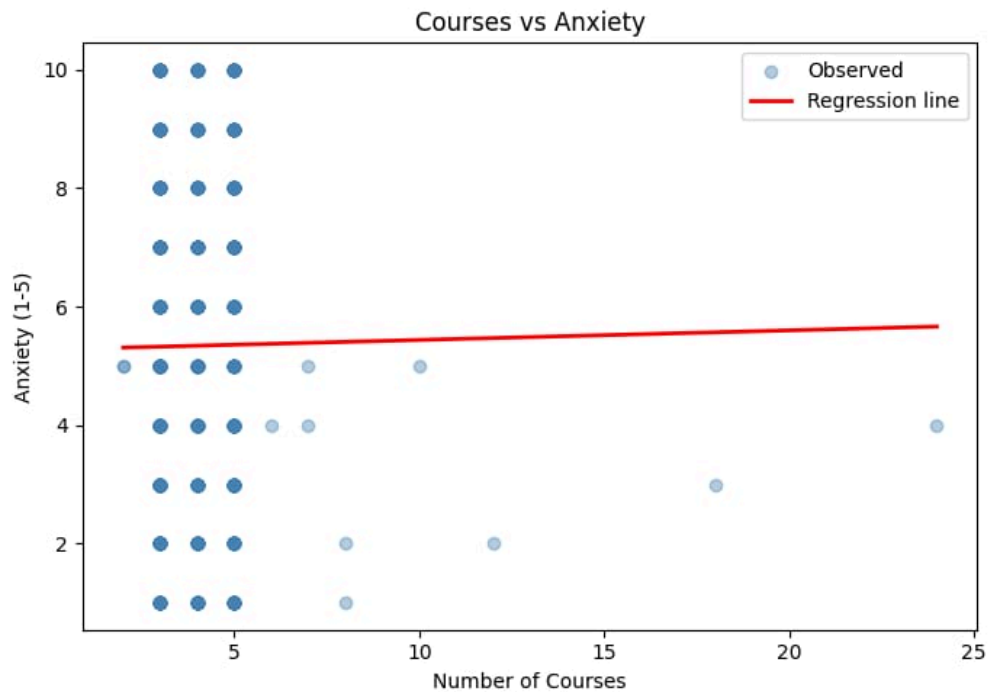
A Pearson correlation matrix was used to evaluate relationships between variables that later helped modeling for regression analysis. The strongest observed relationship is a negative correlation between sleep quality and stress (-0.2429). This can suggest that better sleep quality is associated with reduced stress levels.

A positive correlation between stress and anxiety (0.1778) suggests that these two psychological factors tend to increase together. Study hours exhibit a weak positive correlation with stress (0.0752) and anxiety (0.0488) which indicates that there is minimal effect of academic workload on psychological strain. Sleep hours show a weak positive relationship with sleep quality (0.1355) and weak negative relationships with stress and anxiety, suggesting a slight protective effect of better sleep.

CGPA exhibits very weak correlations with all variables, indicating that academic performance is not strongly influenced by the measured behavioral or psychological factors in this dataset.

7. Linear Regression Analysis :





This report investigates the relationship between weekly study hours, stress level, sleep quality among students using linear regression modeling. The analysis aims to determine the strength and direction of this relationship to better understand how academics affect student well-being.

Linear regression was applied to three models:

Model 1: Study Hours Vs. Stress Level

Model 2: Sleep Hours Vs. Stress Level

Model 3: Courses Vs. Anxiety Level

Correlation coefficients (r), coefficients of determination (R^2), and regression parameters were calculated to assess model fit and predictive utility.

These three relationships were chosen because they represent common student life factors where cause-effect assumptions are often made. While the correlation coefficients are not exceptionally high, they allow for meaningful exploration of weak-to-moderate relationships, reflecting real-world complexity where multiple variables influence stress and sleep.

Model 1: Study Hours vs. Stress Level

Correlation (r): $\approx +0.075$ (Very weak positive relationship)

(From correlation matrix: $r = 0.075161$)

R^2 : 0.0056 (Only 0.56% of variation is explained)

Regression Equation:

$\text{Stress} = b_0 + b_1 \times \text{Study Hours}$

Where, $b_0 = 5.0361$ and $b_1 = 0.02166$

Interpretation:

An increase of 1 hour in weekly study time is associated with an average increase of 0.02166 units in stress level. Although the effect size is extremely small, the slope is statistically significant, indicating that study hours have a measurable linear relationship with stress.

However, the very low R^2 value (0.56%) shows that study hours explain almost none of the variation in stress levels. This means that while the relationship is statistically detectable due to the large sample size, it is not practically meaningful and has very limited predictive power.

Model 2: Sleep Hours vs. Stress Level

Correlation (r): ≈ -0.023 (Very weak negative relationship)

(From correlation matrix: $r = -0.022741$)

R^2 : 0.0005 (Only 0.05% of variation is explained)

Regression Equation:

$\text{Stress} = b_0 + b_1 \times \text{Sleep Hours}$

Where, $b_0 = 5.5939$ and $b_1 = -0.0435$

Interpretation:

An increase of 1 hour of sleep is associated with an average decrease of 0.0435 units in stress level. However, this effect is extremely small and statistically insignificant.

The near-zero R^2 value (0.05%) indicates that sleep hours have virtually no explanatory power in predicting stress levels. This suggests that sleep duration alone is not a reliable predictor of stress in this dataset.

Model 3: Courses vs. Anxiety Level

Correlation (r): $\approx +0.007$ (Extremely weak positive relationship)

(From correlation matrix: $r = 0.007102$)

$R^2: \approx 0.050$ (About 5.0% of variation explained — note: model fit slightly higher than pairwise correlation)

Regression Equation:

$$\text{Anxiety} = b_0 + b_1 \times \text{Courses}$$

Where, $b_0 = 5.2770$ and $b_1 = 0.01614$

Interpretation:

An increase of 1 course is associated with an average increase of 0.01614 units in anxiety level. However, the relationship is extremely weak and statistically insignificant.

Although the R^2 value suggests a small amount of explained variation ($\approx 5\%$), the near-zero correlation and insignificant t-value indicate that this relationship is unreliable. Therefore, the number of courses cannot be considered a meaningful predictor of anxiety in this dataset.

8. Performance Evaluation

To evaluate the performance of each linear regression model, we used R-squared (R^2) and correlation coefficients (r) to assess goodness-of-fit, predictive strength, and model behavior.

Model Performance:

Model	Dependent Variable	Independent Variable	r	R^2	b_1 (Slope)	b_0 (Intercept)
1	Stress Level	Study Hours	+0.075	0.0056	0.0217	5.0361
2	Anxiety Level	Courses	+0.007	0.050	0.0161	5.2770
3	Stress Level	Sleep Hours	-0.023	0.0005	-0.0435	5.5939

Model 1 explains only 0.56% of the variance in stress, indicating extremely weak predictive power. Although the relationship is statistically significant, study hours do not meaningfully explain variations in stress levels.

Model 2 explains approximately 5.0% of the variance in anxiety, making it the strongest among the three models. However, the correlation is nearly zero ($r \approx 0.007$) and statistically insignificant, suggesting that the relationship is weak and unreliable.

Model 3 explains just 0.05% of the variance in stress, demonstrating almost no explanatory power. Sleep hours, when considered independently, are not effective in predicting stress levels.

Overall, all models exhibit very low R^2 values, indicating that the selected independent variables fail to capture the majority of variation in stress and anxiety. This suggests that other unobserved factors play a much more significant role.

9. Discussion

The results reveal that the relationship between academic workload and mental health is much weaker than expected. None of the models show strong predictive power, with the highest R^2 being only 5%.

Study hours and sleep hours have almost no meaningful effect on stress, while courses show only a slight and unreliable influence on anxiety. This indicates that workload alone is not a strong determinant of student mental health.

These findings highlight that stress and anxiety are influenced by more complex and personal factors such as coping ability, lifestyle, and individual perception of pressure. Students with similar workloads often experience very different stress levels, which weakens any consistent statistical relationship.

Overall, the analysis suggests that student well-being cannot be effectively explained using single-variable linear models. Future studies should include multiple variables or more advanced methods to better capture these complex relationships.

10. Conclusion:

This project explored the relationship between academic workload and student mental health using statistical analysis and regression models. Although the relationships were generally weak, the study highlighted the complexity of real-world data and showed that multiple factors influence stress and anxiety. One of the main challenges was handling inconsistent data and interpreting low correlations meaningfully. Through this work, I gained practical experience in data preprocessing, exploratory analysis, and applying statistical methods. Overall, this project reflects the effort and learning developed throughout the Statistics for Data Science course and provides a strong foundation for future data-driven analysis.

Survey: [Link](#)

Dataset:

1. [Original Dataset](#)
2. [Generated Dataset](#)